

FIRST-COLLISION HURWITZ DYNAMICS ON MINIMAL CYCLE FACTORIZATIONS: SHARP TAILS, FIXED CENSUS, AND EXACT FIBRES

ANONYMOUS

ABSTRACT. Let $\mathcal{F}_n$ be the minimal transposition factorizations of the long cycle $(1\,2\,\dots\,n)$. At each epoch, find the first adjacent pair of transpositions having the same lower endpoint and apply the right Hurwitz move there; hold if no such pair exists. Although each local move is classical, this state-dependent scheduler creates a terminating finite map. We prove that successive update positions increase strictly, giving the sharp tail bound $n-2$. Via the classical lower-endpoint bijection with parking functions and Pollak's circular model, the fixed-state count is $(n-1)^{n-2}$. Independently, we classify every predecessor of every labelled target by one inverse Hurwitz test. The maximum one-step fibre is $n-1$, attained uniquely by the canonical adjacent-transposition factorization. Exact exhaustion through $n=8$ checks 280,392 states and reveals a stronger binomial law for complete scheduler histories; because an all- n bijection is not yet proved, that law is stated only as a conjecture. Hurwitz actions, factorization-parking correspondences, and ordinary Prüfer enumerations receive zero contribution credit. The owner gate is `OWNER_RED_AMBER/HOLD_EXTERNAL`.

1. CONVENTION, MAP, AND SUBTRACTION

Permutations compose in the standard order: in $\sigma\rho$, the permutation ρ acts first. Put $c_n = (1\,2\,\dots\,n)$ and

$$(1) \quad \mathcal{F}_n = \{(\tau_1, \dots, \tau_{n-1}) : \tau_i \text{ is a transposition and } \tau_1 \cdots \tau_{n-1} = c_n\}.$$

For a transposition (a, b) written with $a < b$, set $\ell((a, b)) = a$. The right Hurwitz move at i is

$$(2) \quad H_i(\dots, \tau_i, \tau_{i+1}, \dots) = (\dots, \tau_{i+1}, \tau_{i+1}\tau_i\tau_{i+1}, \dots).$$

It preserves the product.

Define $T_n : \mathcal{F}_n \rightarrow \mathcal{F}_n$ by choosing the least $i \in \{1, \dots, n-2\}$ with

$$(3) \quad \ell(\tau_i) = \ell(\tau_{i+1})$$

and applying H_i ; if no such i exists, fix the state. The product, Hurwitz orientation, numeric endpoint order, and least-index convention are part of the definition. Reversing only the long-cycle orientation does not preserve the dynamics.

Minimal long-cycle factorizations, their Cayley-tree count, and Hurwitz actions are classical [1, 2]. The lower-endpoint map

$$(4) \quad (\tau_1, \dots, \tau_{n-1}) \mapsto (\ell(\tau_1), \dots, \ell(\tau_{n-1}))$$

is the classical bijection from $\mathcal{F}_n$ to parking functions of length $n-1$ [3, 4]. All of this receives zero contribution credit. The residual claims concern the adaptive first-collision scheduler and its target-resolved inverse graph.

Internally, direct Hurwitz actions have repeatedly been owner-killed, and P181 uses first-descent prefix reversal on ordinary permutations. The present map is not a braid-group orbit census, a 0-Hecke action, or a prefix reversal. Indeed its conditional local operators fail the braid relation already on $((1, 4), (1, 3), (1, 2))$. A bounded owner search found the classical ingredients above but no literal scheduler theorem package. This non-hit is not evidence of novelty, priority, completeness, or freedom to operate.

2. STRICTLY ADVANCING COLLISIONS

For $f \in \mathcal{F}_n$, let $i_1, i_2, \dots$ be its successive update positions.

Key words and phrases. Hurwitz action, transposition factorization, parking function, finite dynamical system, transient, fibre.

Theorem 2.1 (scheduler normal form and sharp tail). *Every nonempty update history is strictly increasing:*

$$(5) \quad 1 \leq i_1 < i_2 < \dots \leq n-2.$$

Consequently every recurrent state is fixed and every tail is at most $n-2$. The bound is sharp; one witness is

$$(6) \quad ((1, n), (1, 2), (2, 3), \dots, (n-2, n-1)),$$

whose history is $1, 2, \dots, n-2$ and whose terminal state is $((1, 2), (2, 3), \dots, (n-1, n))$.

Proof. At an active position write $\tau_i = (a, b)$ and $\tau_{i+1} = (a, c)$, where $a < b, c$. The two transpositions are distinct: otherwise they cancel and the n -cycle would be expressed using fewer than $n-1$ transpositions. The update replaces them by

$$(7) \quad (a, c), (b, c).$$

The new lower endpoints at positions $i, i+1$ are $a, \min(b, c)$, and the second is strictly larger than a . All earlier lower endpoints are unchanged, including the endpoint at position i . Since i was the first old equality, no equality is created before i and the equality at i disappears. The next update position is therefore strictly larger. This proves (5), termination, and the upper bound.

For (6), the first move changes the initial two factors to $(1, 2), (2, n)$. The next collision is at position two. Inductively the factor (j, n) meets $(j, j+1)$, and the Hurwitz move replaces them by $(j, j+1), (j+1, n)$. The stated history and terminal chain follow. $\square$

The proof supplies a pointwise certificate: the history is a subset of $[n-2]$ written in increasing order, and the depth is its cardinality.

3. EXACT FIXED-STATE CENSUS

Put $N = n-1$. A parking function is a word $a \in [N]^N$ whose increasingly sorted entries $b_1 \leq \dots \leq b_N$ satisfy $b_i \leq i$.

Theorem 3.1 (fixed factorizations). *The fixed states of T_n are exactly the factorizations whose lower-endpoint word has no equal adjacent letters, and*

$$(8) \quad |\text{Fix}(T_n)| = (n-1)^{n-2}.$$

Proof. The characterization is the stopping rule. Under (4), it remains to count parking functions $a_1 \dots a_N$ with $a_i \neq a_{i+1}$ for $i < N$.

Use Pollak's circular parking model on $N+1$ spots [5]. There are $(N+1)N^{N-1}$ circular preference words with adjacent entries unequal: choose the first preference freely and each later one in N ways. Simultaneous translation modulo $N+1$ preserves the inequalities, every translation orbit has size $N+1$, and each orbit has exactly one representative that becomes an ordinary parking function after placing the empty spot at the end. Division by $N+1$ gives N^{N-1} , which is (8). $\square$

4. EVERY-TARGET INVERSE HURWITZ ATLAS

Fix a target $y = (\sigma_1, \dots, \sigma_{n-1})$. Let

$$(9) \quad j(y) = \min\{j : \ell(\sigma_j) = \ell(\sigma_{j+1})\},$$

with sentinel value $j(y) = n-1$ when the set is empty. For $i < j(y)$, write $\sigma_i = (a, b)$ with $a < b$. Call i reverse-admissible when

$$(10) \quad \sigma_{i+1} = (b, c) \text{ for some } c > a,$$

where the transposition is unordered, so the condition means that it contains b and its other endpoint exceeds a .

Theorem 4.1 (complete labelled fibre). *Every target has exact indegree*

$$(11) \quad \text{indeg}(y) = \mathbf{1}_{\{j(y)=n-1\}} + \#\{i < j(y) : i \text{ is reverse-admissible}\}.$$

For each counted i , the unique nonself predecessor is $H_i^{-1}y$. Moreover,

$$(12) \quad \max_{y \in \mathcal{F}_n} \text{indeg}(y) = n-1,$$

and the unique maximizer is the canonical chain

$$(13) \quad ((1, 2), (2, 3), \dots, (n-1, n)).$$

Proof. Suppose a nonself source updates at i to y . Positions before i are unchanged, while the source equality at i is removed, so necessarily $i < j(y)$. Inverting (2) gives the pair

$$(14) \quad (\sigma_i \sigma_{i+1} \sigma_i, \sigma_i).$$

For its two lower endpoints to agree, σ_{i+1} must contain the upper endpoint b of $\sigma_i = (a, b)$, and its other endpoint c must exceed a . The coincident-transposition case would cancel and is impossible. Thus (10) is necessary. Conversely it turns (14) into $(a, c), (a, b)$; positions before i retain no collision, so the scheduler selects exactly i . This proves the atlas and uniqueness of each inverse.

A fixed target contributes itself and at most one inverse at each of the $n-2$ positions; a nonfixed target contributes fewer. Hence $\text{indeg}(y) \leq n-1$. Every adjacent pair of (13) is reverse-admissible, so it attains equality. Equality forces a fixed target with every position admissible. If $\sigma_i = (a_i, b_i)$, admissibility makes both endpoints of σ_{i+1} exceed a_i , whence $a_{i+1} > a_i$. A strictly increasing parking function of length $n-1$ is necessarily $(1, 2, \dots, n-1)$, and the classical lower-endpoint bijection then forces the unique factorization (13). $\square$

Iterating (11) gives an exact finite reverse dynamic program for every higher fibre; no uniform closed form is asserted.

5. A VERIFIED HISTORY LAW LEFT OPEN

The exact controls expose a stronger pattern. We isolate it to prevent finite evidence from silently becoming a theorem.

Conjecture 5.1 (history-set law). *For every $I \subseteq [n-2]$,*

$$(15) \quad \#\{f \in \mathcal{F}_n : \text{Hist}(f) = I\} = (n-1)^{n-2-|I|}.$$

Equivalently, the number of states of exact depth t would be

$$(16) \quad \binom{n-2}{t} (n-1)^{n-2-t}.$$

The verifier proves (15) only for $2 \leq n \leq 8$ by exhaustion. A separate Cayley-tree stream also checks $n = 9$. The histogram matches the occurrence-set law of a distinguished letter in a Prüfer word, but tested standard factorization–tree bijections do not identify the scheduler history with those positions. No all- n bijection is presently known here. Accordingly (15), the resulting unique deepest-state claim, and any derived basin formula receive no theorem status.

6. CONTROLS AND LIMITATIONS

The paper-local standard-library verifier generates the full Hurwitz orbit from (13), checks its Cayley cardinality and product, the lower-endpoint parking bijection, every orbit, every history mask through $n = 8$, and every target fibre against (11). It also checks the orientation-sensitive convention: only the frozen product/order is covered. These computations are regression pressure, not proof or novelty evidence.

The paper does not claim a new Hurwitz action, parking-function bijection, or tree code. It does not cover inverse-cycle conventions, other reflection groups, nonminimal factorizations, random schedulers, or the conjectural all- n history law. External circulation remains unauthorized under `HOLD_EXTERNAL`.

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